STACKS, DEFORMATION THEORY: LEMMA-INDUCED-MAP

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5529-5544 of sources/stacks-defos/pinned-original-source.tex; proof span is lines 5546-5572, the complete author proof (25 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 32 of queue_defos.json (pb=979, stmt 5529-5544, proof 5546-5572).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 05TC: sources/stacks-defos/pinned-original-source.tex, source lines 5572–5572. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\alpha : K \to L$ be a map of $D^-(\mathcal{O})$. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}$-modules. Let $n \in \mathbf{Z}$.*

(1) *If $H^i(\alpha)$ is an isomorphism for $i \geq n$, then $H^i(\alpha \otimes^{\mathbf{L}}_{\mathcal{O}} id_{\mathcal{F}})$ is an isomorphism for $i \geq n$.*
(2) *If $H^i(\alpha)$ is an isomorphism for $i > n$ and surjective for $i = n$, then $H^i(\alpha \otimes^{\mathbf{L}}_{\mathcal{O}} id_{\mathcal{F}})$ is an isomorphism for $i > n$ and surjective for $i = n$.*

Proof. Choose a distinguished triangle

$$K \to L \to C \to K[1]$$

In case (2) we see that $H^i(C) = 0$ for $i \geq n$. Hence $H^i(C \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{F}) = 0$ for $i \geq n$ by (the dual of) Derived Categories, Lemma 05TC. This in turn shows that $H^i(\alpha \otimes^{\mathbf{L}}_{\mathcal{O}} \mathrm{id}_{\mathcal{F}})$ is an isomorphism for $i > n$ and surjective for $i = n$. In case (1) we moreover see that $H^{n-1}(L) \to H^{n-1}(C)$ is surjective. Considering the diagram

$$\begin{array}{ccc} H^{n-1}(L) \otimes_{\mathcal{O}} \mathcal{F} & \longrightarrow & H^{n-1}(C) \otimes_{\mathcal{O}} \mathcal{F} \\ \downarrow & & \| \\ H^{n-1}(L \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{F}) & \longrightarrow & H^{n-1}(C \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{F}) \end{array}$$

we conclude the lower horizontal arrow is surjective. Combined with what was said before this implies that $H^n(\alpha \otimes^{\mathbf{L}}_{\mathcal{O}} \mathrm{id}_{\mathcal{F}})$ is an isomorphism. □

Date: stacks-project:defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.